Calum Murray

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Profile

I am currently a postdoctoral researcher at CosmoStat (CEA Paris-Saclay), working on observational cosmology. My research focuses on understanding how galaxies populate the cosmic web by combining weak gravitational lensing observations with detailed characterization of observational systematics. I develop physically motivated models and forward-modeling strategies validated on real survey data from Euclid, UNIONS, and DESI, with particular focus on cluster cosmology and characterizing the galaxy-halo connection through both galaxy clustering and intrinsic alignments. I hold leadership positions in the Euclid collaboration as co-lead of the cluster catalogue work package and the COMB-CL processing function for cluster weak lensing measurements.

Research experiences

01/09/2024 – Present: Postdoctoral researcher at CosmoStat, CEA Saclay, France
Advisors: Dr. Martin Kilbinger & Dr. Samuel Farrens

01/09/2022 – 01/09/2024: Postdoctoral researcher at AstroParticle and Cosmology laboratory (APC), Université de Paris, France
Advisor: Prof. James G. Bartlett

01/11/2020 – 01/09/2022: Postdoctoral researcher at Laboratory of Subatomic Physics and Cosmology (LPSC), Université Grenoble Alpes, France
Advisor: Dr. Céline Combet

Education and key qualifications

30/10/2020: PhD, Astrophysics, Université Paris Cité, France
Thesis: *Galaxy cluster cosmology with the Euclid Dark Energy Survey*
Supervisor: Prof. James G. Bartlett

2017: Master of Mathematical Physics (MPhys), University of Edinburgh, UK
Thesis: *The colours of galaxies*
Supervisor: Prof. Robert Mann

Leadership roles in major collaborations

Euclid Consortium (2017 – Present)

- **Co-lead** of the Galaxy Clusters work package on catalogue selection and definition (since 2024)
- **Co-lead** of COMB-CL processing function for cluster weak lensing measurements (improved code speed by factor of 50 for large datasets) (since October 2025)
- **Lead author** on key paper analysing covariances between observed cluster properties using the 5,150 square degree Euclid Flagship simulation, in preparation.

UNIONS – Ultraviolet Near Infrared Optical Northern Survey (2024 – Present)

- Member, contributing to cosmic shear analysis and intrinsic alignment measurements using UNIONS galaxy shape measurements combined with DESI spectroscopic data

LSST Dark Energy Science Collaboration (2020 – Present)

- Member, contributed two-halo term computation and elliptical halo validation to DESC cluster weak lensing processing function CLMM

Wide-field Spectroscopic Telescope (WST)

- Member of WST Science Team

Statistical and methodological expertise

- Weak gravitational lensing: shear and magnification measurements, cluster mass estimation
- Development of statistical techniques: Wiener filtering, two-dimensional shear field analysis
- Forward modeling and simulation-based inference approaches
- Cross-correlation analyses for tests of gravity and cosmological systematics
- Extended GPU correlation code (cucount, https://github.com/adematti/cucount) for arbitrary spin-correlations, enabling rapid computation of shape-position and higher-order correlations (100x speedup)
- Cluster abundance cosmology: likelihood development, super-sample covariance
- Large-scale survey data validation and systematic characterization
- Analysis of photometric surveys (Euclid, UNIONS, HSC) and spectroscopic surveys (DESI)
- Production of mock observations and catalogue-level forward models

Supervising & Teaching

PhD Supervision

- Antonin Corinaldi (Sept 2025 – present), intrinsic alignments with UNIONS, DESI and Euclid, co-supervised with Dr. M. Kilbinger and Dr. S. Codis
- Constantin Payerne (co-supervised during his PhD), cluster abundance cosmology, resulted in 2 publications
- Raphaël Kou (co-supervised during his PhD), tests of modified gravity, resulted in 1 publication

Master's Students

- Antonin Corinaldi (Jan–Sept 2025), intrinsic alignment with UNIONS and DESI
- Huy Nguyen Phu (Apr–Sept 2024), galactic acceleration using quasar proper motions, resulted in 1 publication (A&A, accepted)
- Théo Courty (May–July 2023), intrinsic alignments of galaxies
- Mahmoud Osman (Apr–June 2022), optical mass proxies for galaxy clusters
- Margaux Scavini (May–June 2019), strong gravitational lensing

Teaching

- August 2026: Lecturer, Euclid Summer School. Delivered 6-hour graduate-level course on observational aspects of weak gravitational lensing
- August 2025: Lecturer, Euclid Summer School. Delivered 6-hour graduate-level course on observational aspects of weak gravitational lensing
- June 2025: Lecturer, COLOURS Workshop. Developed and taught workshop on analyzing Euclid's first data release

Contributions to the research community

Scientific life

- Since 2026: **Peer reviewer for Springer Nature**
- Since 2025: Organizer of monthly meetings for CEA researchers working on Euclid
- Since 2025: Peer reviewer for American Astronomical Society (AAS) journals
- 2022–2024: Organizer of weekly APC cluster journal club
- 2018–2020: Organizer of biweekly APC cosmology journal club
- 2021: Local Organizing Committee, Euclid France meeting, LPSC Grenoble
- 2019: Organizing Committee, Rencontres des Jeunes Physicien.ne.s, Collège de France, Paris

Outreach activities

- Since Nov 2023: Co-organizer of Astronomy on Tap, Paris. Principal organiser since January 2025. Organized monthly public lectures in bars in Paris making astronomy accessible to the general public
- Nov 2022 & Dec 2024: Rencontres du Ciel et de l'Espace, Paris. Volunteered at Euclid and Rubin stands explaining cosmological surveys to the public

Selected publications

54 papers on arXiv: 5 first-author, 13 co-authored (many with students I co-supervised), 2 conference proceedings and 34 Euclid Collaboration papers; 3 more first-author papers in preparation. Complete list at calumhrmurray.github.io/publications.html.

First-author publications

C. Murray, M. Kilbinger, C. Payerne (2026). Two-dimensional weak lensing shear for cluster mass, concentration, ellipticity and miscentering estimation. arXiv:2609.24948.

C. Murray, R. Kou, J.G. Bartlett (2026). Detecting gravitational lensing by matter currents. *Astronomy & Astrophysics*, 708, A48

C. Murray, C. Combet, C. Payerne, M. Ricci (2025). Filtering out large-scale noise for cluster weak lensing mass estimation. *Astronomy & Astrophysics*, 697, A141

C. Murray, J.G. Bartlett, E. Artis, J.B. Melin (2022). Measuring weak lensing masses on individual clusters. *Monthly Notices of the Royal Astronomical Society*, 512(4), 4785-4791

C. Murray (2022). The effects of lensing by local structures on the dipole of radio source counts. *Monthly Notices of the Royal Astronomical Society*, 510(2), 3098-3101

C. Murray, M. Kilbinger (under internal review within UNIONS). Observational evidence for the origin of intrinsic alignments of galaxies.

C. Murray et al. (in preparation). LSx: a catalogue of 71 million galaxies with SPHEREx spectra and distances.

Euclid Collaboration: **C. Murray**, S. Bhargava, et al. (in preparation). Euclid preparation: Covariances between observed cluster properties with the Euclid Flagship simulation.

Key collaborative publications (co-supervised students)

C. Payerne, **C. Murray**, H. Simon (2026). Simulation-based cosmological inference from optically-selected galaxy clusters with Capish. *Astronomy & Astrophysics* (accepted), arXiv:2602.01911

P.H. Nguyen, **C. Murray** (2026). Observing the peculiar acceleration of our Solar System with quasar proper motions. *Astronomy & Astrophysics* (accepted), arXiv:2605.30367 (from my Master's student's internship)

C. Payerne, **C. Murray**, C. Combet, M. Penna-Lima (2024). Towards including super-sample covariance in the unbinned likelihood for cluster abundance cosmology. *Monthly Notices of the Royal Astronomical Society* (accepted), arXiv:2401.10024

C. Payerne, **C. Murray**, C. Combet, C. Doux, A. Fumagalli, M. Penna-Lima (2023). Testing the accuracy of likelihoods for cluster abundance cosmology. *Monthly Notices of the Royal Astronomical Society*, 520(4), 6223-6236

R. Kou, **C. Murray**, J.G. Bartlett (2024). Constraining f(R) gravity with cross-correlation of galaxies and cosmic microwave background lensing. *Astronomy & Astrophysics*, 686, A193

Other significant collaborative publications

E. Artis, J.B. Melin, J.G. Bartlett, **C. Murray** (2021). Impact of the calibration of the halo mass function on galaxy cluster number count cosmology. *Astronomy & Astrophysics*, 649, A47

R. Akhmetshyn, J. Elvin-Poole, M.J. Hudson, et al., incl. **C. Murray** (2026). Measuring satellite galaxy subhalo masses in redMaPPer clusters with UNIONS weak lensing data. *Monthly Notices of the Royal Astronomical Society* (accepted), arXiv:2607.14207

C. Payerne, C. Yèche, W. d'Assignies Doumerg, H. Hildebrandt, M. Kilbinger, **C. Murray**, et al. (2026). Assessing the large-scale angular clustering of UNIONS Lyman Break Galaxies via cross-correlations. arXiv:2607.09943

M. Aguena, C. Combet, S. Fu, C. Payerne, M. Ricci, et al., incl. **C. Murray** (2026). CLMM v2, the updated LSST-DESC cluster weak lensing mass modelling library. arXiv:2610.37455

C. Payerne, Z. Zhang, M. Aguena, C. Combet, et al., incl. **C. Murray** (2025). Weak lensing mass-richness relation of redMaPPer clusters in the LSST DESC DC2 simulations. *Astronomy & Astrophysics*, 700, A34

Euclid Collaboration publications

Euclid Collaboration: S. Bhargava, C. Benoist, A.H. Gonzalez, ..., **C. Murray**, et al. (2025). Euclid Quick Data Release (Q1): First detections from the galaxy cluster workflow. arXiv:2503.19196
- For this article I contributed creating colour images for validation of the cluster detections, validation of the photometry within crowded fields, and the effects of masking

UNIONS Collaboration publications

F. Hervas Peters, M. Kilbinger, et al., incl. **C. Murray** (2025). UNIONS: a direct measurement of intrinsic alignment with BOSS/eBOSS spectroscopy. *Astronomy & Astrophysics*, 699, A201

Plus 33 further Euclid Collaboration papers and 2 conference proceedings; see the complete list

Invited talks & conference presentations

Selected invited seminars

- July 2025: Le comptage d'amas de galaxies dans notre Univers avec Euclid, 27ème Congrès Général de la Société Française de Physique, Troyes, France
- January 2025: Constraining f(R) gravity with cross-correlation of galaxies and cosmic microwave background lensing, Max Planck Institute for Extraterrestrial Physics, Munich, Germany
- April 2024: Beyond cosmic variance: galaxy cluster weak lensing, Royal Observatory of Edinburgh, United Kingdom
- March 2024: Robust estimation of the quasar dipole, Laboratoire Univers et Particules de Montpellier, France
- February 2023: Gravitational lensing by matter currents, CPPM, Marseille, France
- January 2021: Individual weak lensing masses with HSC, LPSC, Grenoble, France
- January 2020: Individual weak lensing masses with HSC, CosmoStat, CEA Saclay, France
- December 2018: Individual weak lensing masses for Euclid, Observatoire de Paris, Meudon, France

Conference & collaboration presentations

8 international conferences (including Rencontres de Moriond, Tracing Cosmic Evolution with Clusters of Galaxies Sexten, Progress on Old and New Themes in Cosmology Avignon) · 25 Euclid collaboration meetings (Euclid France, general Consortium meetings, Galaxy Clusters Science Working Group, Theory Science Working Group) · 8 LSST Dark Energy Science Collaboration meetings

References

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